

CSA1711 - Artificial Intelligence

CO3 Assessment Tool - 2

MCQ Test - Questions, Options and Answer Key

Q1. Which of the following best describes a Logical Agent in AI?

- a) An agent that reacts randomly to the environment
- b) An agent that uses a knowledge base and logical inference to decide actions ✓ Correct Answer**
- c) An agent that only follows pre-defined scripts
- d) An agent that learns from rewards

Justification: A logical agent maintains a knowledge base of facts/rules and uses logical inference (e.g., forward/backward chaining, resolution) to decide its next action.

Q2. In Propositional Logic, ' $P \rightarrow Q$ ' is logically equivalent to:

- a) $P \wedge \neg Q$
- b) $\neg P \vee Q$
- c) $\neg Q \rightarrow \neg P$ only
- d) Both b and c ✓ Correct Answer**

Justification: $P \rightarrow Q$ is equivalent to its material-implication form $\neg P \vee Q$, and also to its contrapositive $\neg Q \rightarrow \neg P$ — so both b and c are valid equivalent forms.

Q3. Conjunctive Normal Form (CNF) requires the formula to be expressed as:

- a) A disjunction of conjunctions
- b) A conjunction of disjunctions (clauses) ✓ Correct Answer**
- c) A conjunction of literals only
- d) A sequence of implications

Justification: CNF is a conjunction (AND) of clauses, where each clause is a disjunction (OR) of literals.

Q4. In FOL, the sentence $\forall x: \text{Student}(x) \rightarrow \text{Passes}(x)$ means:

- a) Some students pass
- b) Every entity that is a student passes ✓ Correct Answer**
- c) If something passes, it is a student
- d) There exists at least one student who passes

Justification: The universal quantifier $\forall x$ combined with the implication states that for every x , being a Student implies Passes — i.e., every student passes.

Q5. Unification in FOL is the process of finding substitutions that:

- a) Convert clauses to CNF
- b) Make two logical expressions syntactically identical ✓ Correct Answer**
- c) Prove a goal using backward chaining
- d) Derive facts using Modus Ponens

Justification: Unification finds a substitution θ that makes two atomic sentences/terms syntactically identical, enabling rules and facts to be matched.

Q6. In the Resolution algorithm, deriving the empty clause $\square$ indicates:

- a) The knowledge base is consistent
- b) The negated goal is satisfiable
- c) A contradiction — the original goal is proven ✓ Correct Answer**
- d) More clauses need to be added

Justification: Deriving the empty clause means the negated goal, combined with the KB, is unsatisfiable — a contradiction — which proves the original (un-negated) goal is true.

Q7. Forward Chaining is best described as a _____ strategy.

- a) Goal-driven / top-down
- b) Data-driven / bottom-up ✓ Correct Answer**
- c) Heuristic-based depth-first
- d) Probabilistic inference

Justification: Forward chaining starts from known facts (data) and fires applicable rules to derive new facts, moving 'upward' toward conclusions — a data-driven, bottom-up strategy.

Q8. Backward Chaining begins reasoning from:

- a) All known facts toward a conclusion
- b) The goal and works backward to verify supporting facts ✓ Correct Answer**
- c) A random clause in the knowledge base
- d) The resolution refutation tree

Justification: Backward chaining starts from the goal, recursively breaking it into sub-goals until each sub-goal is matched against known facts.

Q9. Which step is NOT part of converting a sentence to CNF?

- a) Eliminate biconditionals ($\leftrightarrow$)
- b) Move negations inward using De Morgan's laws
- c) Apply Skolemization to remove existential quantifiers ✓ Correct Answer**
- d) Distribute AND over OR

Justification: The core CNF conversion process (eliminate implications/biconditionals $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute OR over AND) applies to propositional logic. Skolemization is a separate, FOL-specific technique used only when existential quantifiers are present — it is not a step of the basic CNF conversion algorithm.

Q10. Knowledge Engineering in FOL involves:

- a) Training a neural network on domain-specific data
- b) Defining domain concepts, relations, and axioms formally in FOL ✓ Correct Answer**
- c) Writing heuristic search strategies
- d) Encoding reward functions for reinforcement learning

Justification: Knowledge engineering is the process of formally identifying and encoding a domain's objects, relations, and axioms into a First Order Logic knowledge base.

Q11. Modus Ponens states: If $P \rightarrow Q$ is known and P is true, then:

- a) $\neg Q$ is true
- b) Q must be true ✓ Correct Answer**
- c) P is uncertain
- d) The KB is inconsistent

Justification: *Modus Ponens: from $P \rightarrow Q$ and P , we validly infer Q .*

Q12. The Most General Unifier (MGU) is the substitution that:

- a) Is the most specific possible match
- b) Makes two terms identical with minimal variable binding ✓ Correct Answer**
- c) Eliminates all variables from a clause
- d) Applies resolution to produce CNF

Justification: *The MGU is the least-constraining substitution that unifies two expressions, binding variables only as much as necessary.*

Q13. Which inference method is more appropriate when a specific goal is known?

- a) Forward Chaining
- b) Truth Table method
- c) Backward Chaining ✓ Correct Answer**
- d) A* search

Justification: *Backward chaining is goal-driven, working backward from a known goal to check supporting facts, making it efficient when a specific goal is to be verified.*

Q14. Propositional Logic differs from First Order Logic because it:

- a) Supports probabilistic reasoning
- b) Cannot represent objects, relations, or quantifiers — only truth values of atomic propositions ✓ Correct Answer**
- c) Is always more expressive
- d) Handles uncertainty better

Justification: *Propositional logic only deals with whole atomic propositions being true/false; it lacks FOL's ability to represent objects, relations, functions, and quantifiers.*

Q15. Resolution Refutation proves a goal G by:

- a) Directly deriving G from the KB using forward chaining
- b) Adding $\neg G$ to the KB and deriving a contradiction (empty clause $\square$) ✓ Correct Answer**
- c) Checking G in the truth table
- d) Applying Modus Tollens repeatedly

Justification: *Resolution refutation negates the goal, adds it to the CNF knowledge base, and resolves clauses until the empty clause (contradiction) is derived, proving the original goal.*

Q16. Learning from observation in AI refers to:

- a) An agent receiving numerical rewards from the environment
- b) Improving performance by examining examples or experience from the environment ✓ Correct Answer**
- c) Acquiring knowledge by following explicit symbolic rules
- d) Memorizing a fixed dataset without generalization

Justification: Learning from observation means an agent improves its behaviour/performance by examining examples or past experience, rather than being explicitly programmed.

Q17. Inductive learning is best described as:

- a) Deriving specific conclusions from general rules
- b) Generalizing a hypothesis from specific training examples ✓ Correct Answer**
- c) Memorizing training data exactly
- d) Applying search algorithms to find optimal policies

Justification: Inductive learning builds a general hypothesis/model by generalizing patterns observed in specific labeled training examples.

Q18. In a Decision Tree, what does each internal node represent?

- a) A class label / output decision
- b) A probability distribution over outcomes
- c) A test on an attribute / feature ✓ Correct Answer**
- d) A training data sample

Justification: Internal nodes of a decision tree represent a test/split on a particular attribute; leaf nodes hold the class labels/decisions.

Q19. Explanation-Based Learning (EBL) differs from inductive learning because it:

- a) Requires large amounts of training data to generalize
- b) Learns only from numeric data using regression
- c) Uses prior domain knowledge to explain and generalize a single example ✓ Correct Answer**
- d) Applies unsupervised clustering to find hidden patterns

Justification: EBL uses existing domain knowledge to construct an explanation of why one example is a member of a concept, then generalizes that explanation — needing far fewer examples than pure induction.

Q20. In statistical learning, a Naive Bayes classifier assumes:

- a) All features are conditionally dependent on each other given the class
- b) The data must follow a uniform distribution
- c) All features are conditionally independent given the class label ✓ Correct Answer**
- d) Labels are determined by a reward function

Justification: The 'naive' assumption in Naive Bayes is that all input features are conditionally independent of one another given the class label.

Q21. Learning with complete data (no hidden variables) in a Bayesian network involves:

- a) Using the EM algorithm to estimate missing values

- b) Training a deep neural network with dropout regularization
- c) Directly computing maximum likelihood estimates from observed data ✓ Correct Answer**
- d) Applying reinforcement signals to update weights

Justification: When all variables are observed (complete data), parameters can be estimated directly via maximum likelihood from the observed frequencies, with no need for EM.

Q22. The Expectation-Maximization (EM) algorithm is used when:

- a) Training data is fully labeled and no variables are hidden
- b) An agent explores an unknown environment using trial and error
- c) The dataset contains hidden (latent) variables that cannot be directly observed ✓ Correct Answer**
- d) Backpropagation fails to converge in a neural network

Justification: EM iteratively estimates parameters and hidden/latent variable values when some data is unobserved, alternating between an Expectation step and a Maximization step.

Q23. In a Neural Network, the activation function serves to:

- a) Initialize the weights of each neuron to zero
- b) Normalize the input dataset before training
- c) Introduce non-linearity to enable learning of complex patterns ✓ Correct Answer**
- d) Store previously learned examples in memory

Justification: Activation functions (e.g., ReLU, sigmoid) introduce non-linearity, allowing neural networks to model complex, non-linear relationships rather than just linear combinations.

Q24. Which of the following correctly describes backpropagation in a neural network?

- a) Weights are updated in the forward pass using input gradients
- b) Outputs of each layer are stored and reused in later epochs
- c) The error is propagated backward from output to input layers to adjust weights ✓ Correct Answer**
- d) A separate model is trained for each hidden layer independently

Justification: Backpropagation computes the output error and propagates it backward through the network layers, using the chain rule to compute gradients and update weights.

Q25. A Support Vector Machine (SVM) — a type of kernel machine — classifies data by:

- a) Minimizing the number of training examples used
- b) Grouping similar data points into clusters using centroids
- c) Finding the maximum-margin hyperplane that separates classes ✓ Correct Answer**
- d) Updating weights using Q-values from an environment

Justification: SVMs find the hyperplane that maximizes the margin between classes, using the closest points (support vectors) to define that margin.

Q26. The kernel trick in kernel machines (SVMs) allows:

- a) Reducing the number of support vectors needed
- b) Accelerating gradient descent convergence
- c) Operating in a high-dimensional feature space without explicitly computing coordinates ✓ Correct Answer**

- d) Replacing hidden layers in a deep neural network

Justification: The kernel trick computes inner products in a high-dimensional feature space implicitly (via a kernel function), avoiding the cost of explicitly transforming data into that space.

Q27. In Reinforcement Learning, the agent aims to maximize:

- a) Classification accuracy on a fixed test set
- b) The number of features selected from the training data
- c) Cumulative reward received over time through interaction with the environment ✓ Correct Answer**
- d) The size of the neural network hidden layer

Justification: RL agents learn a policy that maximizes the expected cumulative (long-term) reward obtained through interacting with the environment.

Q28. The Q-Learning algorithm in Reinforcement Learning is classified as:

- a) A supervised learning algorithm requiring labeled datasets
- b) An unsupervised clustering technique using centroids
- c) A model-free, off-policy temporal difference learning method ✓ Correct Answer**
- d) A decision-tree-based ensemble classifier

Justification: Q-Learning learns action-values directly from experience without a model of the environment (model-free) and learns the optimal policy regardless of the policy being followed (off-policy), using temporal-difference updates.

Q29. Overfitting in machine learning occurs when:

- a) The model performs poorly on both training and test data
- b) The learning rate is set too low during gradient descent
- c) The model fits training data too closely and generalizes poorly to new data ✓ Correct Answer**
- d) The training set has more features than training examples

Justification: Overfitting happens when a model captures noise/specific patterns in the training data too closely, harming its ability to generalize to unseen data.

Q30. Which of the following best distinguishes Reinforcement Learning from Supervised Learning?

- a) RL uses labeled input-output pairs; SL uses reward signals
- b) Both RL and SL require explicit reward functions
- c) RL learns via trial-and-error with reward feedback; SL learns from labeled examples ✓ Correct Answer**
- d) RL is always faster to converge than SL on any dataset

Justification: RL learns a policy through trial-and-error interaction with an environment guided by reward signals, whereas SL learns a mapping from labeled input-output example pairs.
